

AgriConnect – Offline-First Farmer Marketplace & Advisory Platform

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ABSTRACT

AgriConnect – Offline-First Farmer Marketplace & Advisory Platform

- AgriConnect is an offline-first platform designed to support farmers in areas with poor or unreliable internet connectivity. It provides **offline access to complete crop information**, including planting, fertilizers, irrigation, diseases, pests, harvesting, storage, and selling guidance. The platform also enables farmers to manage products, interact with buyers, and view buyer trust ratings. **Market prices and marketplace data are synchronized and updated whenever internet connectivity is available**, while essential information remains accessible offline.



INTRODUCTION

- Agriculture plays an important role in the livelihood of millions of people in India. However, farmers in rural and remote areas often face poor or unavailable internet connectivity, making it difficult to access digital agricultural services.
- **AgriConnect is an offline-first farmer marketplace and agricultural advisory platform designed to work in no-network and low-network conditions.** It provides farmers with offline access to useful crop information, including planting, irrigation, fertilizers, diseases, pests, harvesting, storage, and selling guidance.
- The platform also allows farmers to manage products, view buyer information, check trust ratings, and manage orders. **Market prices and other dynamic information are updated whenever internet connectivity becomes available.**
- Important information and offline actions are stored locally and **synchronized with the backend server when connectivity is restored**, helping farmers continue essential activities with reduced dependence on continuous internet connectivity.



EXISTING SYSTEM

- Most existing agricultural platforms depend on continuous internet connectivity.
- Farmers may lose access to **crop guidance, disease information, fertilizers, market prices, and marketplace services** in low-network areas.
- Product listings and order updates may fail or be delayed when connectivity is poor.
- Existing systems provide limited support for **offline agricultural knowledge and data synchronization**.
- Therefore, they are less effective for farmers in rural areas with unreliable internet connectivity.



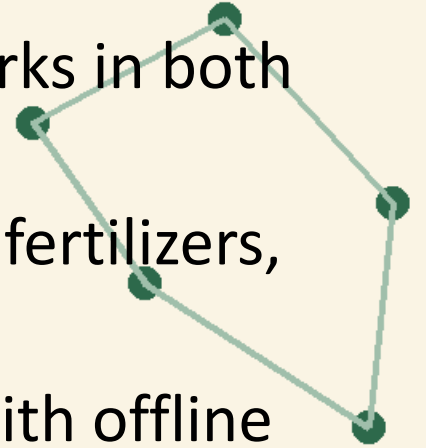
PROBLEM

- Farmers in rural areas often struggle with unreliable internet connectivity, making it difficult to access agricultural guidance, market prices, and online marketplaces. There is a need for a platform that keeps **essential farming knowledge available offline** while synchronizing **dynamic market and marketplace information** when connectivity is restored. AgriConnect addresses this gap by combining **offline agricultural advisory, market information, and a connectivity-resilient farmer marketplace.**



SOLUTION(Contribution)

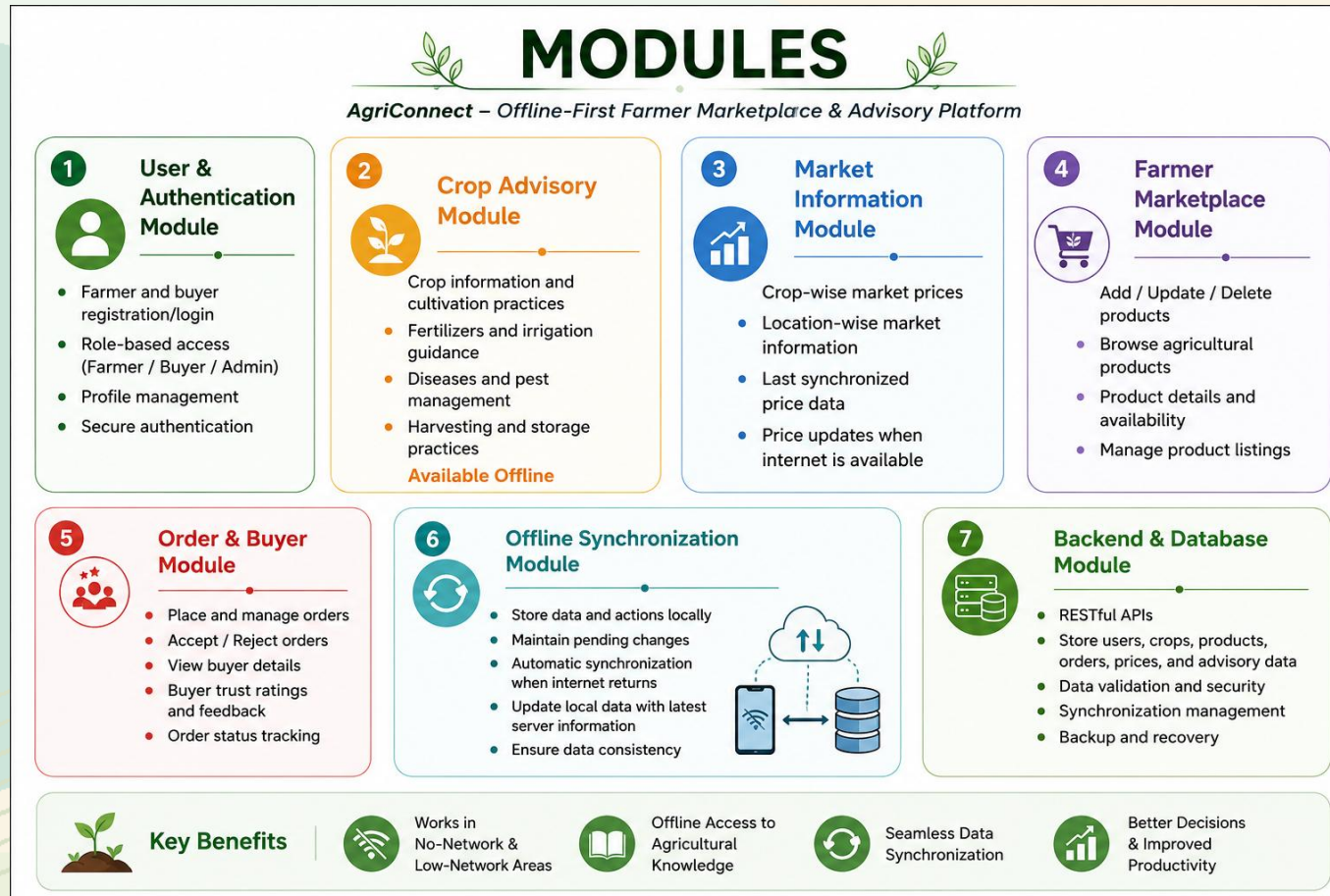
- **AgriConnect** provides an offline-first agricultural platform that works in both low-network and no-network conditions.
- It provides **offline access to crop information**, including planting, fertilizers, irrigation, diseases, pests, harvesting, and storage.
- Farmers can **manage products, orders, and buyer information**, with offline actions stored locally.
- **Market prices and other dynamic data** are updated when internet connectivity becomes available.
- The system automatically **synchronizes locally stored data with the backend server** when the network is restored.
- This helps farmers access essential agricultural knowledge and make better selling decisions **without depending on continuous internet connectivity**.



Objectives

- To develop an **offline-first agricultural platform** for farmers in low or no-network areas.
- To provide **offline access to comprehensive crop information**, including cultivation, fertilizers, diseases, pests, harvesting, and storage.
- To enable farmers to **manage products, buyers, and orders** with reduced dependency on internet connectivity.
- To provide **updated market prices and location-based market information** whenever internet connectivity is available.
- To implement **automatic synchronization** between locally stored data and the backend server.
- To help farmers make **better farming and selling decisions** through accessible agricultural and market information.

Modules



SOFTWARE REQUIREMENTS

OS: Windows/Linux

Frontend: HTML, CSS, JavaScript

Backend: Java + Spring Boot

Database: MySQL

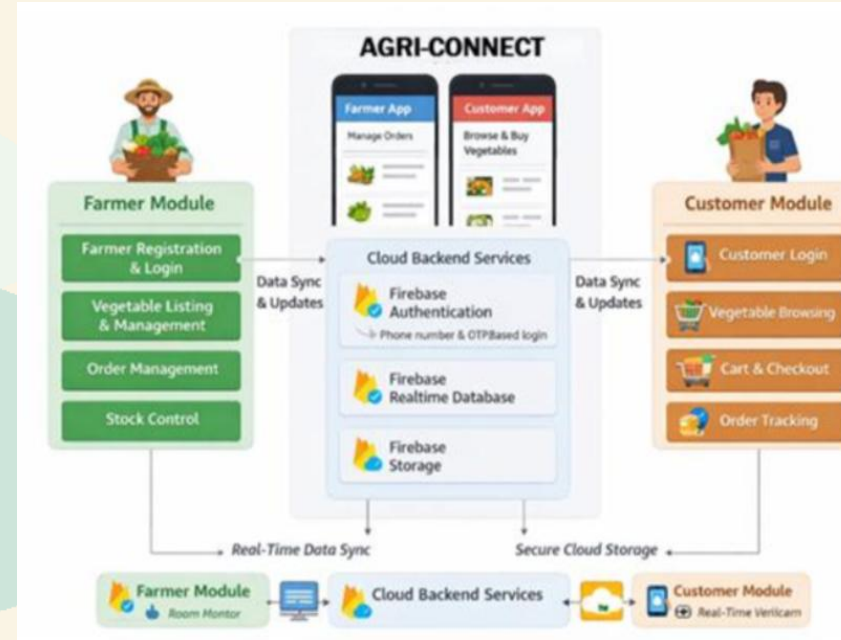
API Testing: Postman

IDE: IntelliJ IDEA / VS Code

Version Control: Git & GitHub

Build Tool: Maven

Offline Storage: Local Storage / Indexed DB



CONCLUSION AND FUTURE SCOPE

AgriConnect provides an offline-first platform that gives farmers access to essential crop knowledge, marketplace services, and synchronized market information even in low-connectivity areas.

§ It helps farmers make **better farming and selling decisions** while reducing dependency on continuous internet connectivity.

§ **Future Scope:** AI-based disease detection, weather and market-price prediction, regional language/voice support, IoT-based crop monitoring, and integration with government agricultural services.

THANK YOU

